one month, floods the next. "Climate change is loading the dice by increasing the duration, frequency, scale and intensity of extreme weather events," says Murtugudde.

WHAT INDIA NEEDS TO DO

To combat climate change and limit its damage, the Modi government has already set ambitious targets: achieving net zero emissions by 2070, reducing carbon dioxide emissions by a billion tonnes by 2030 and meeting half of India's energy needs through renewables in the same period. Meanwhile, faced with the relentless rise in extreme weather events, experts and government agencies are increasingly turning to science and technology for better prediction, early warning and mapping of vulnerabilities. IMD has done commendable work in cyclone forecasting, saving countless lives. But even as its seasonal forecasts, which cover entire monsoon periods, are improving, experts want them to get hyperlocal and at scale so that citywide prevention measures can be planned in advance. Likewise, decadal predictions for the next 10–20 years, which estimate whether a monsoon will be in the up or down phase, should ideally come at the local, if not hyperlocal, level.

Granular risk maps are another key mitigation measure. The current ones outline the hazards each district is prone to, along with people and assets at risk, besides vulnerability maps to assess communities' ability to cope with the hazards. But experts want them developed at much higher resolutions, on kilometre scale, for each potential threat, from drought, heat wave, fire and rain to floods. Equally important is a constant update of these profiles, as rapid climate change adds new risks to existing ones. Risks, therefore, need to be reassessed yearly to ensure they are factored into development planning, says Ramesh.

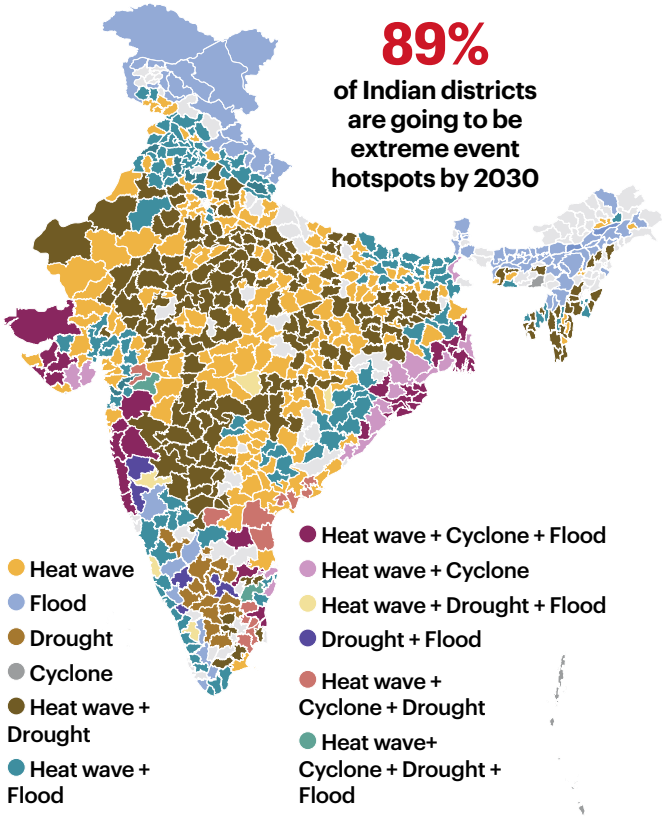
Several states have rolled out action plans to combat heat waves, floods, lightning and thunderstorms and build resilience. But resilience on paper is not translating into action on ground. Coordination between central, state and local bodies remains weak and implementation often falters. Sanjay Srivastava, chairman of the Climate Resilient Observing Systems Promotion Council (CROPC), points out how Punjab was warned about the risk of floods as early as April, yet it had no plan of action. "The mindset that nothing will happen is the biggest socio-behavioural problem in the country, " he says. "The culture of safety must be built on a war footing among local bodies and citizens. Even if early warnings are issued, the problem lies in implementation by local authorities and compliance by people."

Mitigation measures such as urban planning regulations for construction, buildings, drainage and power supply must be designed to minimise the impact of disasters when they strike. It remains a huge challenge. Local governance bodies often have knowledge of no-build zones and vulnerable areas, yet grant approvals for unsafe construction or turn a blind eye to violations of safety norms in connivance with local stakeholders. "The biggest problem is the lack of implementation of environmental bylaws by the local bodies and compliance by the common people. This is possible due to the connivance among local government bodies, the contractors, and political patronage. That needs to stop," says Srivastava. Cities, too, must be redesigned. Experts talk of "urban sponges" that include parks, permeable pavements, rain gardens and drainage systems that soak up rainfall instead of letting it deluge streets. "The idea is to ensure falling water walks and walking water stops," says Murtugudde.

Equally critical is strengthening the state disaster response forces (SDRFs). Their primary role is to carry out relief and rescue operations during disasters, but on non-disaster days, their mandate includes educating people, spreading awareness on disaster management and preparing communities with evacuation strategies, alternative shelters and rescue routes. States like UP, West Bengal and Bihar have improved their preparedness through their SDRFs and even conducted mock exercises with citizens in flood-prone areas. However, there should be

THE THREAT AHEAD

By 2030, districts predominantly in coastal areas will be exposed to all four hazards—flood, drought, cyclone and extreme heat—while those in hills and plains will brace for floods, drought and heat wave



Source: State of Extreme Events; Mohanty et al 2024 | IPE Global and Esri India Climate Risk Observatory